Replication of the orthogonal latent-mean contrasts in Jeffries (2024)

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Abstract

Do the latent-mean contrasts in Jeffries (2024) sit on the printed scale −0.40, −0.20, 0, and do the rejection rates in Tables 4.1 and 4.3 follow from that model? Jeffries defines three codes for four groups, states that imbalance makes the set non-orthogonal, and reports multigroup and MIMIC averages near those three numbers when the generating means are $1, 0.6, 0.8, 0.4$. The fitting was done in lavaan 0.6.17. Population contrasts were obtained by fitting each candidate identification to the population moments. A 400-replicate Monte Carlo then used the analysis string from the public OSF project. On a population in which every loading is 0.70, a marker loading of 1 produces −0.28, −0.14, and 0. On the population lavaan builds from the generation string, the first loading is 1 and the other three are 0.70, and the same marker model produces −0.40, −0.20, and 0. The second population matches the printed averages. At $N = 240$, all 30 compared averages fall within two combined Monte Carlo standard errors of the page. The unbalanced power rates 49.23, 21.08, and 7.77 do as well. Six of the 30 rejection rates miss that uncorrected band. Bonferroni and Holm, on the same 30 rates, leave one miss: the equal-group MIMIC null rate 2.26 against the printed 5.18. The dissertation's percents are single Wald tests at $\alpha = 0.05$. The correction applies only to this agreement check. The multigroup convergence rate in the smallest groups is 88.25% against the printed 89.29%.

1 What this replication did

This section answers what was fit, and in which program. lavaan 0.6.17 is the estimator named in the dissertation, and it is the program that produced every fitted number below. simsem 0.5.17, the installed successor of the named simsem 0.5.16, was used to parse the generation string. OpenMx and MplusAutomation are not installed. Mplus is not installed. The file `simulation/mgcfa_osf.inp` mirrors the multigroup string and was not executed. `exportData(program = "Mplus")` in the generation scripts writes data files. The estimation scripts call `lavaan::sem`.

Lean is used only for the code identities already proved in `LatentContrasts/Orthogonality.lean`: the products of the Table 2.2 codes, the population values of equations (3.3)–(3.5), and the treatment coding of those codes. The gap between −0.28 and −0.40 is a question about which population and which identification lavaan is fitting. That question is answered by the moment fits, not by a new proof.

The moment fits are `simulation/candidate_readings.R`. The short Monte Carlo is `simulation/validate_gaps.R`, 400 replicates, seed 13131 under R's L'Ecuyer generator, with one drawn seed per cell. The 10,000-replicate rerun is `simulation/validate_10k.R`. Each of its ten cells has a fixed seed, set once, and the replicates are drawn in order on one core. The agreement classification, including the Bonferroni and Holm checks, is `simulation/score_gaps.R`. Package

versions are in `simulation/out/session_info.txt`: R 4.3.3, lavaan 0.6.17, simsem 0.5.17. The dissertation names R 4.2.2.

Jeffries studies one factor with four indicators and four groups. The true-difference means are $\alpha = (1, 0.6, 0.8, 0.4)$. The null means are all 1. Total sample size is 240, 480, or 960. The group shares are 25/25/25/25, 35/35/15/15, and 45/45/5/5.

2 Comparison

Table 1 puts one published claim on each row. "Match" means the printed value and our value agree as exact numbers, as the same population target, or, for a Monte Carlo rate, as a gap no larger than two combined standard errors. That rate rule is one comparison at a time. The combined standard error uses the printed count of 10,000 and our count of converged replicates. "Partial" means one piece holds and another does not. "No match" means the reading is on a different scale. The last two rows are the agreement check on all 30 rates, uncorrected and then corrected. The dissertation's own $\alpha = 0.05$ columns are single tests. They are not re-thresholded here.

Claim	Published	Ours	Verdict
Unweighted products of the Table 2.2 codes	Equation (2.4) is zero	Proved zero	match
Contrasts on $(1, 3/5, 4/5, 2/5)$	$-0.4,\ -0.2,\ 0$	$-2/5,\ -1/5,\ 0$	match
Third contrast while the means differ	Used as a null contrast	Proved equal to 0	match
Imbalance destroys orthogonality	The set becomes non-orthogonal	Only code 1 with code 3 fails	partial
OSF marker model, generation string as lavaan parses it	Averages near $-0.40,\ -0.20,\ 0$	Population fit $-0.40,\ -0.20,\ 0$	match
Every loading 0.70, then marker loading 1	The same averages	Population fit $-0.28,\ -0.14,\ 0$	no match
Table 4.1, $N = 240$, shares 45/45/5/5, percent significant	49.23, 21.08, 7.77	48.16, 25.21, 7.08	match
Table 4.3, same cell, percent significant	8.13, 8.59, 7.73	10.80, 12.85, 9.77	partial
Table 4.5, same cell, multigroup convergence	89.29%	88.25% of 400	match
Thirty rejection rates, two combined standard errors	Each printed percent is one test at $\alpha = 0.05$	6 of 30 outside the band	partial
Same 30 rates, Bonferroni $z = 2.94$ and Holm	Those tests are a single-test design	1 of 30 outside the band	partial

Table 1: Published claims against this replication. Monte Carlo rows use 400 replicates at $N = 240$. The population fits use the moment matrices, not a sample. The last two rows correct only the agreement check.

The product row is equation (2.4), proved at 103. The population row is equations (3.3)–(3.5) on the printed means, proved at 123. The third contrast is zero while the first two means differ, proved at 127. The imbalance row is partial because 68 shows that sizes (a, a, b, b) leave two of the three weighted products at zero and set the other to $2(b - a)$. At $N = 240$ that product is -96 in condition 2 and -192 in condition 3.

The two scale rows are the gap that used to read as a no match for the whole table. They are different populations. Section 3 scores every reading against the same printed target. The

percent rows use the population that matches −0.40. The old comparison of 7.77 with 4.5 was the every-loading-0.70 marker run, whose z statistics sit on the rescaled contrasts. That run is not the row in Table 1.

3 Where the gaps come from

The question is which population and which identification produce −0.40, −0.20, and 0, and whether that same fit reproduces the printed rejection rates.

The generation string in the public OSF script, and the repaired form of the appendix string, is

```
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~ 1*1
eta ~ c(1, .6, .8, .4)*1
```

with residual variances 0.51. lavaan 0.6.17 keeps a free loading on `i1` and drops the second modifier 0.7. `simulateData` fills the free loading with 1. A draw of 20,000 at factor mean 1 had mean 1.996 on `i1` and 1.693 on `i2`. The targets are 2 and 1.7. The variances were 1.021 and 0.749. The targets are 1.01 and 0.755. simsem 0.5.17 rejects the page's token `i1 ~ 1*` with the same parse error. The OSF generation script already writes `1*1`. The dissertation page writes `1*` for that one intercept. Figure 3 is the page.

The analysis string in `osf/analysis/mgcfa_model.R` is a marker loading of 1, free factor variances, four free means, and one intercept shared by all four indicators and all four groups. It does not constrain the loadings equal across groups, and it does not fix them at 0.7. The summary script sets the printed standard-error column to the standard deviation of the estimates divided by $\sqrt{10{,}000}$.

3.1 Parameter ledger

Status is what the source supports. "Printed" is on the page or in the public script. "Inferred" is a consequence of fitting that string in lavaan 0.6.17. "Assumed" is a choice this replay still had to make.

Parameter	Status	Value and source
Loadings in the Chapter III sentence	printed	0.70 for all four indicators. PDF page 89.
Factor variance, residual, intercept	printed	0.50, 0.51, and 1. The 49% sentence is $1 - \lambda^2$ and needs a unit factor variance. The appendix numbers do not.
Latent means	printed	$(1, 0.6, 0.8, 0.4)$ and, in the null, all 1.
Table 2.2 codes	printed	$C_1 = (-1, 1, -1, 1)$, $C_2 = (-1, -1, 1, 1)$, $C_3 = (1, -1, -1, 1)$.
Appendix and OSF code `c3`	printed	The opposite sign. Groups 1 and 4 are -1. The population value is 0 either way.
Marker sentence	printed	First loading fixed at 1, intercepts equated, latent variances free. PDF page 90.
Analysis string	printed	OSF `mgcfa_model.R`. Marker 1, one common intercept, four free means.
First loading under the generation string	inferred	Free, and filled with 1. Loadings 2–4 stay at 0.7. `out/appendix_lavaanify.csv`.
`i1 ~ 1*` on the PDF page	printed	Does not parse in lavaan 0.6.17 or simsem 0.5.17.
`i1 ~ 1*1` in the OSF script	printed	The token this replay generates from.
Standard-error column	inferred	`sd(estimate) / sqrt(n)` in the OSF summary script.
Significance	printed	Two-sided $p < 0.05$ from the MLR parameter table.
Software versions	printed	lavaan 0.6.17, simsem 0.5.16, R 4.2.2, MLR, 10,000 iterations. This replay uses lavaan 0.6.17, simsem 0.5.17, R 4.3.3, and `iter.max` 10,000.
Random stream	assumed	L'Ecuyer, `set.seed(13131)`, then one seed per cell. Not the simsem stream.
MIMIC latent intercept	inferred	The posted line `eta ~ c1 + c2 + c3` fixes it at 0. Adding an intercept matches the printed averages.

3.2 Each reading against the printed averages

Table 2 scores each reading against the printed target -0.40, -0.20, 0. Moment rows are the lavaan fit to the population mean vector and covariance, with 10^7 observations per group, so Monte Carlo error is not the comparison. MIMIC rows are one sample of 100,000 per group, seed 7, except the `contr.sum` row, which is the 20,000-per-group sample from seed 13131.

Two different constraints land on -0.40: fixing the measurement at the equal-loading population, and leaving the first loading at the value lavaan generates from `NA*i1`. The public analysis script is the second constraint. It is a marker model. On the equal-loading population that script returns -0.28. On the parsed generation string it returns -0.40. The printed tables match the second of those. The earlier replay generated every loading at 0.70 and then fixed the first at 1, which is the first row of Table 2. Its Monte Carlo averages, -0.286, -0.134, and 0.007 at 200 replicates, are that row and not a failure of the optimizer.

A free latent intercept is required for the MIMIC averages. The line in `osf/analysis/mimic_model.R` does not include one, and lavaan 0.6.17 then fixes the latent intercept at 0. The large-sample slopes are about -0.15, 0.06, and 0.27. The saved summary is near -0.40. The fit that adds the intercept lands on -0.399, -0.200, and 0.000. The posted loop also assigns the z statistic back onto the estimate before the row is stored. The saved summary contains values near -0.40, so that summary was not produced by that assignment.

Reading	Ours	Explanatory parameter	Verdict
All loadings 0.70, marker, first intercept 0	−0.280, −0.140, 0	Marker rescales by 0.7	no match
All loadings 0.70, OSF common-intercept marker	−0.280, −0.140, 0	Same rescaling	no match
All loadings 0.70, loadings and intercepts fixed	−0.400, −0.200, 0	Measurement fixed at the population	match
All loadings 0.70, factor variance fixed at 0.5, first loading free	−0.400, −0.200, 0	Scale set by ψ	match
Parsed generation string, OSF marker	−0.400, −0.200, 0	First loading generated at 1	match
Parsed generation string, marker, first intercept 0	−0.400, −0.200, 0	A common shift of the means	match
Parsed string, group-1 mean fixed at 0	−0.276, −0.115, −0.038	Reference mean under a shared intercept	no match
Parsed string, loadings fixed at 0.7	−0.431, −0.215, −0.001	The first loading is not 0.7	no match
Factor variance 1, loadings 0.70, marker	−0.280, −0.140, 0	Marker still rescales by 0.7	no match
Wishart likelihood versus normal, OSF model	Difference below 10^{-13}	n against $n-1$	no move
ML versus MLR, one large sample	Difference 0	Robust standard errors	no move
MIMIC, latent intercept free, factor codes	−0.399, −0.200, 0.000	Treatment contrast equals the defined contrast	match
MIMIC, posted line, intercept fixed at 0	−0.150, 0.062, 0.273	The fixed intercept	no match
MIMIC, numeric codes ±1, intercept free	−0.208, −0.097, 0.001	The code changes by 2	no match
MIMIC, numeric effects codes ±1	Half the contrast, sign follows the code	Code direction	no match
MIMIC, `contr.sum` assigned to the factors	Same slope as treatment coding	lavaan did not switch the slope	no move

Table 2: Candidate readings against the printed averages −0.40, −0.20, 0. The printed column is that target on every row.

3.3 Monte Carlo noise

The question is whether the remaining percent gaps are the size of a 400-replicate simulation. A gap is outside noise when it exceeds two combined standard errors. For a percent, the combined standard error is

$$100\sqrt{\frac{\hat{p}(1-\hat{p})}{10{,}000}+\frac{\tilde{p}(1-\tilde{p})}{n}},$$

with $\hat{p}$ the printed rate and n our number of converged replicates. For an average, the printed standard error is already the standard error of the mean of 10,000 replicates, and ours is the standard deviation divided by $\sqrt{n}$.

All 30 averages in the $N = 240$ cells we ran, multigroup and free-intercept MIMIC, true and null, are inside that band. The sampling standard deviations match the printed standard errors times 100. In the multigroup 45/45/5/5 true-difference cell the printed standard errors are all 0.0021, so the implied standard deviations are 0.210. Ours were 0.210, 0.221, and 0.206.

The 7.77 against 4.5 comparison does not survive this population. The 4.5 was the third contrast's rejection rate in the equal-loading marker run. Here the same printed rate is 7.77 and ours is 7.08. The combined standard error is 1.39 points.

Printed cell, $N = 240$	Ours	Verdict
Table 4.1, equal groups, averages -0.4046, -0.2027, -0.0010	-0.412, -0.204, 0.004	within noise
Table 4.1, shares 45/45/5/5, averages -0.4038, -0.2181, 0.0054	-0.394, -0.222, 0.011	within noise
Table 4.1, that cell, percent significant 49.23, 21.08, 7.77	48.16, 25.21, 7.08	within noise
Table 4.3, that cell, percent significant 8.13, 8.59, 7.73	10.80, 12.85, 9.77	partial
Table 4.5, that cell, convergence	353/400 = 88.25% against 89.29%	within noise

Table 3: The cells that motivated the gap check. Multigroup model, parsed generation population, 400 attempted replicates.

Six of the 30 rejection rates are outside the uncorrected band. All six are percents. Their averages are inside it. The other 24 rates are inside the same band. $|z|$ is the absolute difference divided by the combined standard error.

Cell	Contrast	Printed	Ours	$\lvert z\rvert$	Corrected
Multigroup, equal, true difference	3	5.29	3.27	2.19	inside
Multigroup, 35/15, null	2	5.15	9.02	2.67	inside
Multigroup, 45/5, null	2	8.59	12.85	2.48	inside
MIMIC, 45/5, true difference	1	62.98	57.89	2.02	inside
MIMIC, 45/5, true difference	2	23.37	18.55	2.42	inside
MIMIC, equal, null	1	5.18	2.26	3.77	outside

The corrected column is the same under three rules: Bonferroni at $z = \Phi^{-1}(1-0.05/30) = 2.94$, the two-sided Bonferroni threshold $z = \Phi^{-1}(1-0.05/60) = 3.14$, and Holm. One rate stays outside. It is the MIMIC model, equal groups, null means, contrast 1: 2.26 against 5.18, $z = -3.77$, two-sided $p = 0.00016$, from 399 converged replicates.

3.4 Multiplicity

The percent columns in Tables 4.1–4.4 are one Wald test on one dataset at $\alpha = 0.05$. That is the design of those columns. This replication leaves those tests at $\alpha = 0.05$. Multiplicity is a property of the agreement check only: thirty differences between a printed rate and our rate.

The uncorrected check is a two-sided 5% test, $|z| > 1.96$. The two-combined-standard-error band is that rule with 2 in place of 1.96. On these thirty rates the two cutoffs select the same six rows.

Bonferroni divides 0.05 by 30. The one-sided normal quantile of the remaining tail is 2.94. The two-sided quantile is 3.14. Holm sorts the thirty two-sided p-values and compares the ith with $0.05/(31 - i)$, then stops at the first p-value that fails its cutoff. The smallest p-value is 0.00016, below $0.05/30 = 0.00167$, so Holm rejects that row. The next p-value is 0.0076, from the multigroup 35/15 null contrast 2. Holm's second cutoff is $0.05/29 = 0.0017$. The step-down stops. Five of the six uncorrected misses are inside every corrected rule above.

A binomial test asks whether six misses in thirty trials exceeds the count expected at $p = 0.05$. The upper tail is 0.0033. Under independence the expected count is 1.5. The thirty indicators are dependent. Each cell contributes three contrasts computed on the same replicates, so a miss on one contrast moves with the other two. Dependence of that kind widens the distribution of the miss

count relative to a Bin(30, 0.05) draw. The binomial probability is the independent-trials figure. The ten cells use separate seeds, so the dependence sits inside cells. Table 4.3's partial mark in Table 1 is the uncorrected miss on contrast 2, 12.85 against 8.59, $|z| = 2.48$. That row is inside the corrected band.

The multigroup null model at $N = 240$ and shares 45/45/5/5 converged in 389 of 400 replicates, 97.25%, against Table 4.5's 95.75%. The combined standard error is about 0.8 points.

4 Assumptions

4.1 Random-number stream

Jeffries sets `set.seed(1313)` and passes seed 13131 to `exportData`. We drew normal factor scores and normal residuals in R with the L'Ecuyer generator and `set.seed(13131)`, then one seed per cell. The replicates are not Jeffries's replicates. The `.dat` files are not in the OSF archives.

4.2 Package versions

The dissertation names R 4.2.2 and simsem 0.5.16. We used R 4.3.3 and simsem 0.5.17. lavaan is 0.6.17 in both. The optimizer is `nlminb`. A replicate counts as converged when `lavInspect(fit, "converged")` is true, which is the flag the OSF script reads from the summary header, and the three contrast standard errors are finite.

4.3 Replicate count

The tables use 10,000 converged replicates. The classification above uses 400 attempts per cell. That length separates a rate of 7.8% from a rate of 4.5%. It leaves a wide standard error on our side of each comparison. The rerun at the printed length is `simulation/validate_10k.R`, ten cells, 10,000 attempts, one fixed L'Ecuyer seed per cell. The seeds are in `simulation/out/seeds_10k.csv`.

4.4 The first intercept

The PDF page writes `i1 ~ 1*`. The OSF generation script writes `i1 ~ 1*1`, as do the other three indicators and the null model on the page. We generated from `1*1`.

5 Ambiguities that were resolved

5.1 The ink matches the text layer

The question was whether a sign or a printed average in the text extract disagreed with the page. On the pages checked here, it does not. Figure 2 has the minus signs on −0.40, −0.20, and 0.

The crops are `paper/figures/`. The script is `paper/figures/crop_pages.py`. It runs `pdftoppm -png -r 600`. Pages whose rotation flag is 90 degrees are turned clockwise into reading order. The command is `python3formalization/latent-contrasts/paper/figures/crop_pages.py`.

5.2 The public analysis script

The question was which identification produced Tables 4.1 and 4.3. The OSF project https://osf.io/q9yu8/ contains the multigroup and MIMIC scripts and the summary files. The multigroup

Latent variables were identified using the marker method approach, where the first indicator's loading was fixed to one ($\lambda_1 = 1$). Per recommendations from Hancock (1997), over both SEM forms, intercepts were modeled such that $\tau_{1_1} = \tau_{2_1} = \ldots \tau_{x_i}$. Further, the variance of all latent variables were freely estimated.

Figure 1: Marker identification. The first loading is fixed at 1, and the intercepts are equated. PDF page 90.

model results for models of data that were explicitly generated to show latent mean differences (see Tables 4.1 and 4.2). As a reminder, the data-generated latent mean difference for Contrasts 1 through 3 were, on average, -0.40, -0.20, and 0 (i.e., a null difference), respectively. In the null model, the data-generated mean differences for Contrasts 1 through 3 were zero, on average, across all three contrasts. An average

Figure 2: The data-generating contrasts are -0.40, -0.20, and 0. PDF page 116.

summary matches Table 4.1 to the printed digits, including 8,723 significant replicates of contrast 1 at $N = 240$ in the equal true-difference cell. The standard-error column in that file is the standard deviation divided by $\sqrt{10{,}000}$.

The artifact is `osf/analysis/mgcfa_model.R`, retrieved 2026-10-27. The moment fit of that string on the parsed generation population is the fifth row of Table 2. The Monte Carlo is `simulation/out/validate_gaps.csv`, from `Rscriptvalidate_gaps.R400out`.

5.3 The third code and the numeric codes

Appendix A and the OSF generation script code `c3` with the opposite sign from Table 2.2. On the printed means the third contrast is zero, so the sign does not move a two-sided rejection rate for that contrast. Numeric codes ± 1 estimate half the defined contrast, because the code changes by 2. Factor codes with a free latent intercept estimate the full contrast. Theorem 159 is the 0/1 form of that identity. The marker theorem at 131 is the every-loading-0.70 reading, and it is not the parsed generation population.

6 What remains unresolved

At 400 replicates, six rejection rates sit outside two combined standard errors. After Bonferroni and after Holm, one of those six remains outside: MIMIC, equal groups, null means, contrast 1, 2.26 against 5.18. The averages in every compared cell are inside the uncorrected band. What was fit is the OSF multigroup string, the free-intercept MIMIC string, MLR, $p < 0.05$, `iter.max` of 10,000, and the appendix factor codes. Jeffries's random stream was not refit. The `.dat` files are not on OSF. Whether the six rates, and the one corrected rate, stay outside at 10,000 replicates is the rerun in `simulation/validate_10k.R`.

```
# ---- Models of N = 240 ----
pacman::p_load(here, tidyverse, simsem, lavaan, purrr)
set.seed(1313)
# ---- Significance Model ----
## ---- Model Syntax ----
# 4 item model
pop_model_cond1 <- "
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~~ 0.51*i1
i1 ~ 1*
i2 ~~ 0.51*i2
i2 ~ 1*1
i3 ~~ 0.51*i3
```

Figure 3: Appendix A. The first intercept is written `1*`. The loading line contains both `NA*i1` and `0.7*i1`.

The page prints −0.4001 for the MIMIC average of contrast 1 at $N = 240$ in the equal true-difference cell. The public summary stores −0.399094. The significance count in that cell is 9,573 on the page and in the file. The other MIMIC cells we checked round from the file onto the page.

The PDF generation line `i1 ~ 1*` does not parse. The OSF script's line `i1 ~ 1*1` does. We did not run simsem 0.5.16. One uncommented `exportData` call, in the $N = 480$ condition-3 null script, names group sizes $(168, 168, 72, 72)$, which is the 35/15 split. Some commented calls name sizes from a different condition than the file name. The datasets those calls would have written are not in the archive. The Monte Carlo in Table 3 uses the printed shares, and the condition-3 averages and the 89.29% convergence rate agree with the page.

Jeffries and Bovaird (2026), DOI `10.1177/01466216261452255`, was not obtained as a PDF. Unpaywall, Europe PMC, and the Wayback machine had no PDF on 2026-10-27. The OSF node is the analysis record for that title. It does not contain the article.

7 Code and source availability

Place	Date	Result
Appendix A of the dissertation PDF	2026-10-27	Generation syntax. sha256 of the PDF is `a21df98c...fd469b3`
OSF `/v2/search/` for the title	2026-10-27	HTTP 404
OSF node `q9yu8` by title filter	2026-10-27	Public project, analysis scripts and summaries, no `.dat` files and no article PDF
DOI `10.1177/01466216261452255`	2026-10-27	No PDF from the publisher page, Unpaywall, Europe PMC, or Wayback
lavaan 0.6.17	2026-10-27	Installed and used for every fit
simsem 0.5.17	2026-10-27	Installed. Rejects `i1 ~ 1*`. The named version is 0.5.16
OpenMx, MplusAutomation, Mplus	2026-10-27	Not installed

The dissertation file was retrieved from the Wayback `id_` object dated 2025-05-11, snapshot `20250511025154`. The help request for a script, filed before this OSF node was found, is https://

github.com/alectionary/vibe-math/issues/33. The fetch request for the 2026 article is https://github.com/alectionary/vibe-math/issues/34.

8 Next steps

The open numerical question is the rejection-rate agreement at the printed length of 10,000 replicates. The script is `simulation/validate_10k.R` and the seeds are `simulation/out/seeds_10k.csv`. That rerun does not replace the missing `.dat` files. simsem 0.5.16, if obtained, would test whether that version fills the free loading with 1 in the same way lavaan 0.6.17 does. The 2026 PDF would show whether the journal version uses these tables.

9 What we proved

The codes, in group order $(1, 2, 3, 4)$, are

$$c_1 = (-1, 1, -1, 1), \qquad c_2 = (-1, -1, 1, 1), \qquad c_3 = (1, -1, -1, 1).$$

The weighted product is $\sum_{g=1}^{4} n_g c_{jg} c_{kg}$. For equal sizes the three products are zero (107). For sizes (a, a, b, b) the products of c_1 with c_2 and of c_2 with c_3 stay zero, and the product of c_1 with c_3 equals $2(b - a)$ (68). The column sums are zero (99).

A defined contrast is the average of the means coded $+1$, minus the average of the means coded -1. On $\kappa = (1, 3/5, 4/5, 2/5)$ the three contrasts are $-2/5$, $-1/5$, and 0 (123). The third is zero while $\kappa_1 \neq \kappa_2$ (127). Multiplying every mean by $7/10$ multiplies the contrasts by $7/10$ (131). That factor is the every-loading-0.70 marker reading. It is not the population lavaan builds from `NA*i1`. The treatment-contrast form with coefficients $-2/5$, $-1/5$, and 0 reproduces κ (159).

The proofs are rational arithmetic. The `ring` proofs depend on `propext`, `Classical.choice`, and `Quot.sound`. The `native_decide` proofs also depend on `Lean.ofReduceBool`. There is no `sorry`.

10 Artifacts

- Code identities. File `LatentContrasts/Orthogonality.lean`. Command: `lakebuild`.
- Moment fits of every reading. File `simulation/out/candidate_readings.csv`. Command: `Rscriptcandidate_readings.Rout`. Seed 13131 for the simulated samples. Moment fits do not use a seed.
- Large-sample MIMIC. File `simulation/out/mimic_large.csv`. Command: `Rscriptmimic_large.Rout`. Seed 7.
- Monte Carlo. File `simulation/out/validate_gaps.csv`. Command: `Rscriptvalidate_gaps.R400out`.
- Agreement classification, 400 replicates. File `simulation/out/gap_scores.csv`, with the family summary in `simulation/out/gap_scores_multiplicity.csv`. Command: `Rscriptscore_gaps.Rout/validate_gaps.csvout/gap_scores.csv`.
- Ten-thousand-replicate rerun. Script `simulation/validate_10k.R`. Seeds: `simulation/out/seeds_10k.csv`. Command: `Rscriptvalidate_10k.R10000out`.

- Equal-loading replay, kept so the old averages stay reproducible. File `simulation/out/monte_carlo.csv`. Command: `Rscriptmonte_carlo.R200100out`.
- Parser checks. File `simulation/out/parser_checks.csv`.
- Session. File `simulation/out/session_info.txt`. Command: `Rscriptsession_info.Rout`.
- OSF scripts and summaries. Directory `osf/`.
- Page crops. Directory `paper/figures/`.

The dissertation PDF is local and gitignored. Its sha256 is `a21df98c20401491e6b4a1d00a21b6f7cee5551a00c04a89fb414ec6cfd469b3`.

Table 4.5
Initial Model Convergence Rates by Condition & Model

N	Condition	MIMIC Frequency Converged (%)	MGCFA Frequency Converged (%)
Significance Models			
240	1	10000 (100%)	9886 (98.86%)
	2	10000 (100%)	9811 (98.11%)
	3	10000 (100%)	8929 (89.29%)
480	1	10000 (100%)	9989 (99.89%)
	2	10000 (100%)	9994 (99.4%)
	3	10000 (100%)	9805 (98.05%)
960	1	10000 (100%)	10000 (100%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	9986 (99.86%)
Null Models			
240	1	10000 (100%)	9932 (99.32%)
	2	10000 (100%)	9911 (99.11%)
	3	10000 (100%)	9675 (95.75%)
480	1	10000 (100%)	9998 (99.98%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	9999 (99.99%)
960	1	10000 (100%)	10000 (100%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	10000 (100%)

Figure 4: Table 4.5. At $N = 240$ and condition 3 the multigroup significance model converged in 8,929 of 10,000 replicates.